# Kanak Garg

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## PROFILE

Software Engineer with a backend focus and hands-on experience in Python, Go, FastAPI, and PostgreSQL, backed by a solid foundation in data structures and algorithms. Built scalable REST APIs, authentication workflows, and database-driven applications. Seeking a software or backend engineering role.

## EDUCATION

**Master's in Computer Science** 2025 – Present
MIT World Peace University, Pune CGPA: 7.5

**Bachelor's in Computer Science** 2022 – 2025
Nowrosjee Wadia College, Pune CGPA: 8.50

## SKILLS

- **Languages:** Python, Go, Java, JavaScript, SQL
- **Core CS:** Data Structures & Algorithms (practice repo), OOP, DBMS
- **Backend & Frameworks:** FastAPI, Django, Node.js, REST API Design, JWT Auth, RBAC
- **Databases:** PostgreSQL, MongoDB, Firebase Firestore, Oracle SQL
- **Cloud & DevOps:** Docker, Google Cloud (Foundations)
- **Tools & Libraries:** Git, GitHub, pytest, pandas, NumPy, scikit-learn

## PROFESSIONAL EXPERIENCE

**Backend Developer** Jul 2025 – Jul 2026
*MIT World Peace University – Project GAIA* *Pune, India*

- Developed 100+ REST API endpoints backed by PostgreSQL for multiple application features.
- Implemented authentication flows and email services, and integrated an optimization framework that improved performance by 110%.
- Strengthened existing APIs with robust request validation and error handling.

## PROJECTS

**BioVote — Biometric Voter Verification System**

- Built a Go (Gin) REST API with 6 endpoints in a team hackathon project for the ECI: Aadhaar + iris verification, booth dashboard, and tamper-evident audit logs to prevent duplicate voting.
- Used Firestore transactions and atomic counters across 4 collections to stop race-condition double votes; implemented JWT auth with role-based access control.
- Protected PII with salted SHA-256 hashing of Aadhaar and iris data, restricted Firestore to backend-only access via security rules, and containerized the service with a multi-stage Docker build.

**Expense Voucher Management System**

- Built a full-stack voucher approval system (React, Node.js/Express, PostgreSQL) with JWT authentication and role-based access control for 3 roles (Employee, Director, Accounts), enforced on both frontend and backend.
- Designed the PostgreSQL schema with CHECK constraints, a sequence for race-free voucher numbers, and a status-history audit trail; used transactions with row locking to prevent duplicate approvals.
- Wrote 22 integration tests with Jest and Supertest against a real test database; added Zod validation, bcrypt password hashing, and search, filter, and pagination APIs.

**Toxic Comment Flagger — Machine Learning**

- Built a machine learning system to classify comments into toxic, obscene, insult, and sarcastic categories.
- Applied TF-IDF features and a neural network (Dense, ReLU, Sigmoid) trained with backpropagation.
- Developed an end-to-end prediction pipeline covering preprocessing, training, and evaluation.

## CERTIFICATIONS

Google Cloud Foundations | Oracle SQL Database | IBM Data Science